

Introduction to Information Systems

1. Why Information Systems Are Important

Information Systems are now an **essential part of society and business**. Almost every organization, whether small or large, uses computers and information systems to help operate its business.

Because computers have become more affordable and powerful, businesses use them to:

- Manage daily operations
- Store and process data
- Communicate with customers
- Monitor business activities
- Support decision-making

As future business and Information Systems professionals, you need to understand **how Information Systems work and how they support an organization**.

Professor's explanation:

Think of an Information System as a tool that helps a business turn **data into useful information** so people can make better decisions.

2. Data and Information

What is Data?

Data refers to raw facts, figures, numbers, characters, or symbols that can be stored and processed by a computer.

For example:

100, 250, 300, 450

These are simply numbers. By themselves, we may not know what they represent.

Data can come from many sources, such as:

- Customer transactions
- Research
- Sales records

- Employee records
- Product information

What is Information?

Information is data that has been **processed, organized, and given meaning**.

For example:

Data:

100, 250, 300, 450

Information:

"The company's sales increased from 100 units to 450 units during the month."

Now the numbers have meaning and can help a manager make a decision.

The important idea is:

Data = Raw facts

Information = Processed and meaningful data

Effective Information and Records Management helps organizations reduce costs, improve efficiency, maximize information assets, and take advantage of new technologies.

Professor's Question

Question: If you receive a list of raw customer purchase numbers without names, dates, or categories, is it data or information?

Answer: It is **data** because it is still raw and lacks context.

To turn it into information, we could analyze it to determine:

- Which product sells the most?
- When do customers usually buy?
- Who are the regular customers?

That information can then support business decisions.

3. Characteristics of Good Information

Good information must have certain characteristics.

1. Accurate

The information must be **correct and reliable**.

If the information is wrong, the decision based on it may also be wrong.

2. Complete

The information should contain **all important details** needed to make a decision.

Incomplete information can lead to incorrect assumptions.

3. Cost-Effective

The cost of collecting and processing the information should be reasonable compared with its value.

In simple terms:

The information should be worth the cost of obtaining it.

4. User-Focused

Information should be appropriate for the person using it.

For example, a CEO may need strategic information, while an employee may need detailed operational information.

5. Relevant

The information must be **related to the problem or decision** being considered.

There is no point giving a manager information that has nothing to do with the current business problem.

6. Authoritative

Information should come from a **reliable and credible source**.

7. Timely

Information must be available **at the right time**.

Even accurate information can become useless if it arrives too late.

Example

Suppose a manager receives this report:

"Many customers are buying our products."

Is this good information?

Not really.

Why?

Because we don't know:

- How many customers?
- Which products?
- Which branch?
- When did they buy?
- How much did they purchase?

Therefore, the information is not sufficiently **complete and accurate** for making an important business decision.

4. System Concepts

A **system** is a group of **interrelated components** that work together to achieve a **common goal**.

A system generally follows this pattern:

Input → Processing → Output

Input

Something enters the system.

Processing

The system transforms or processes the input.

Output

The system produces the result.

Example: Fast-Food Restaurant

Imagine ordering food at a fast-food restaurant.

Input:

- Your order

- Your payment

Processing:

- The restaurant receives your order.
- The kitchen prepares the food.
- The food is packaged.

Output:

- Your completed meal.

The components work together to achieve the common goal: **providing the customer with the requested food.**

5. Components of a System

There are three major components:

Input

Input refers to the elements that enter the system for processing.

Examples:

- Raw materials
- Energy
- Data

Processing

Processing is the **transformation of the input** into something useful.

Output

Output is the **final result** produced by the system.

Example: Creating a PowerPoint Presentation

Input:

- Research
- Text

- Pictures
- Data

Processing:

- Typing
- Editing
- Designing
- Organizing the slides

Output:

- The completed PowerPoint presentation.

Remember:

INPUT → PROCESSING → OUTPUT

This is one of the most important basic concepts in Information Systems.

6. Types of Systems

According to the material, there are four types:

A. Dynamic System

A **Dynamic System** consists of interconnected components that interact with one another and are controlled by management.

In simple terms, the system is **active and continuously changing**.

B. Cyber Native System

A **Cyber Native System** is a Dynamic System that uses **feedback and control**.

Feedback allows the system to determine what is happening and respond accordingly.

C. Open System

An **Open System** interacts with its environment by exchanging inputs and outputs.

For example, a business receives:

Input:

- Customers
- Money
- Information
- Materials

And produces:

Output:

- Products
- Services
- Information

D. Adaptive System

An **Adaptive System** can change itself or respond to changes in its environment in order to survive and succeed.

Example: Grab

A ride-hailing application such as Grab can be viewed as an **Open System** because it receives inputs such as passenger bookings and locations and produces outputs such as driver assignments and completed trips.

It can also demonstrate feedback and adaptation through features such as ratings and pricing adjustments.

7. Information System

An **Information System (IS)** is a group of interconnected components that work together to:

1. Collect data
2. Process data
3. Store information
4. Provide information
5. Support decision-making

An Information System is **not simply a computer program**.

It involves the interaction of **people, processes, organizations, and technology**.

The material discusses three major dimensions of an Information System:

Organization + Management + Technology

8. Organization Dimension

The **Organization Dimension** refers to how the Information System fits into the organization itself.

An Information System contains and supports things such as:

- Functional specialties
- Business processes
- Organizational culture
- Political interest groups
- Standard operating procedures (SOPs)

This is important because every organization operates differently.

For example, the Information System needed by a **bank** will not be exactly the same as the Information System needed by a **restaurant**.

Why?

Because they have different:

- Goals
- Processes
- Employees
- Customers
- Organizational cultures
- Rules and procedures

Therefore, a company cannot simply **copy another company's Information System and expect it to work perfectly**.

9. Management Dimension

The **Management Dimension** focuses on how managers use Information Systems to solve business problems and make decisions.

Managers use information to:

- Allocate resources
- Coordinate activities
- Monitor operations
- Make decisions
- Develop products and services
- Create long-term strategies

Example

Imagine you manage a restaurant and suddenly receive many online orders.

The management side asks:

"What business problem are we facing, and what should we do about it?"

You may decide to:

- Change the workflow
- Add employees
- Improve delivery processes
- Use a better ordering system

The manager is responsible for **identifying the problem and deciding what action to take**.

10. Technology Dimension

The **Technology Dimension** refers to the technology used to support the organization and management.

It includes:

- Computer hardware
- Software

- Data management technology
- Networking and telecommunications

For example, a restaurant may use an integrated **ordering and POS system** connected to delivery applications.

The technology can:

- Automatically receive orders
- Reduce human errors
- Update records
- Speed up service

So remember:

Management decides what needs to be done.

Technology helps make it possible.

11. Transaction Processing System (TPS)

A **Transaction Processing System (TPS)** processes data generated from everyday business transactions.

Examples include:

- Sales transactions
- Payroll
- Billing
- Inventory/stock control

The main purpose of a TPS is to **record transactions, update records, and produce reports.**

Example: Convenience Store

Suppose you buy a bottled water.

The cashier scans the barcode.

The TPS records:

Product sold → Payment received → Inventory updated

The system may automatically reduce the stock count from:

50 bottles → 49 bottles

This is an example of transaction processing.

12. Management Information System (MIS)

A **Management Information System (MIS)** takes data, often from a **Transaction Processing System**, and converts it into **summarized information and reports**.

MIS is commonly used by:

- Middle managers
- Operational supervisors

Examples of MIS reports include:

- Summary reports
- On-demand reports
- Ad-hoc reports
- Exception reports

Examples of MIS applications include:

- Sales management systems
- Human Resource Management Systems

Simple Example

A TPS records thousands of individual sales transactions.

The MIS can summarize them into:

"Total sales this month: ₱500,000."

The manager doesn't need to manually examine every transaction.

13. Decision Support System (DSS)

A **Decision Support System (DSS)** helps users make decisions, especially when the situation is **semi-structured or unstructured**.

Unlike a basic report, DSS can provide:

- Information
- Models
- Data analysis tools
- Alternative options
- Interactive decision-making support

The user is more actively involved in using a DSS than in using a typical MIS.

Example: Bank Loan

Imagine you are a bank manager.

A customer has an unusual financial situation, and you must decide whether to approve the loan.

A **DSS** would be appropriate because you may need to:

- Examine financial data
- Compare alternatives
- Analyze risks
- Consider different scenarios
- Make a judgment

Easy way to remember:

MIS = Reports

DSS = Analysis + Decision Support

14. Expert Systems

An **Expert System** is a knowledge-based Information System designed to help users solve problems or diagnose situations.

It is based on concepts from **Artificial Intelligence (AI)**.

An Expert System contains:

- **A Knowledge Base**

- **Software modules**
- **Inference mechanisms**

Its purpose is to use stored expert knowledge to provide advice similar to an expert consultant.

Simple Example

Imagine a system where you enter symptoms such as:

Fever + cough + sore throat

The system analyzes the information using its knowledge and provides possible recommendations.

The important idea is that the system uses **stored knowledge and inference** to help solve a problem.

★ BSIS Professor's Quick Review

Concept	Simple Meaning
Data	Raw facts
Information	Processed, meaningful data
System	Components working together toward a goal
Input	What enters the system
Processing	Transformation of the input
Output	Result produced by the system
Dynamic System	Interacting and changing system
Cyber Native System	Uses feedback and control
Open System	Exchanges input/output with its environment
Adaptive System	Adjusts to changes to survive
Organization Dimension	How IS fits the organization
Management	How managers use IS to solve problems

Dimension

Technology Dimension	Hardware, software, data, networks
TPS	Processes daily transactions
MIS	Produces summarized management reports
DSS	Helps analyze options and make decisions
Expert System	Uses expert knowledge to help solve problems

The most important chain to remember:

Data → Processing → Information → Decision

And for systems:

Input → Processing → Output

And for Information Systems:

Organization + Management + Technology

If you understand these three relationships, you already have a strong foundation for your **BSIS Information Systems** subject.